

I Built a Search Agent, Then Spent Months Teaching It When Not to Answer.

Wajdi Ben Saad

20/05/2026

Paris Machine Learning Meetup



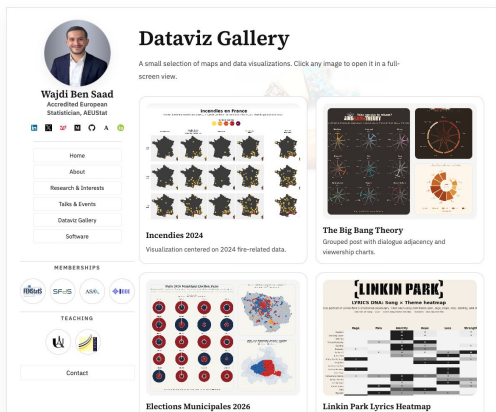
About me



Statistician
Data “scientist”
(Re)searcher

Small Language Models
Efficient training & inference
RL & **Agentic** systems

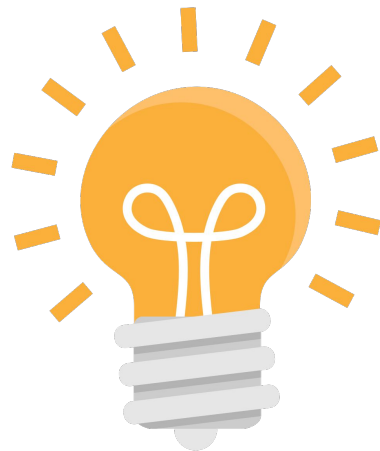
& I teach Data Visualization @ Université Paris Cité



The main lesson today :

a **good** search agent is not the one that **always answers...**

- It is the one that knows when it has **enough evidence**,
- when it needs **a tool**,
- and when it should stay **silent**.





**this person does not exist ;)*

**A KNOW IT ALL IS SOMEONE WHO
KNOWS EVERYTHING**

EXCEPT HOW ANNOYING HE IS



How did the fast food chain establish a significant presence in the United States?



In-N-Out Burger

In - N - Out Burger is an American regional chain of fast food restaurants with locations primarily in the American Southwest and Pacific coast . It was founded in Baldwin Park ,...

Jack in the Box

Jack in the Box is an American fast - food restaurant chain founded February 21 , 1951 , by Robert O. Peterson in San Diego , California , where it is headquartered . The

Big Boy Restaurants

By 1979 there were more than a thousand Big Boy restaurants in the U.S. and Canada , and about 20 franchisees . Shoney 's , Elias Brothers and Frisch 's -- charter franchisees ...

Chipotle Mexican Grill

In a list of fastest - growing restaurant chains in 2009 , Chipotle was ranked eighth , based on increases in U.S. sales over the past year , and in 2010 Chipotle was ranked...

underspecified

There are many fast food chains you could be talking about, like In-N-Out Burger, Jack in the Box, Big Boy Restaurants, or Chipotle Mexican Grill. Which one are you referring to?



Where was the Friends TV show filmed?

Friends was filmed at Warner Bros. Studios in Burbank, California, and the show was never shot in New York. The producers believed that filming outside the studio made episodes less funny and that the live audience was an integral part of the series.



answerable



my friend works at the studio. he said part of it was filmed in new york



According to the information provided, Friends was never filmed in New York.



How can I see my text usage with *PhoneCo*

To see your text usage with *PhoneCo*, you can sign in to *PhoneCo* website and view your usage details on the Usage page. From there, you can choose to view or save a printable bill in PDF format. ...

answerable

...

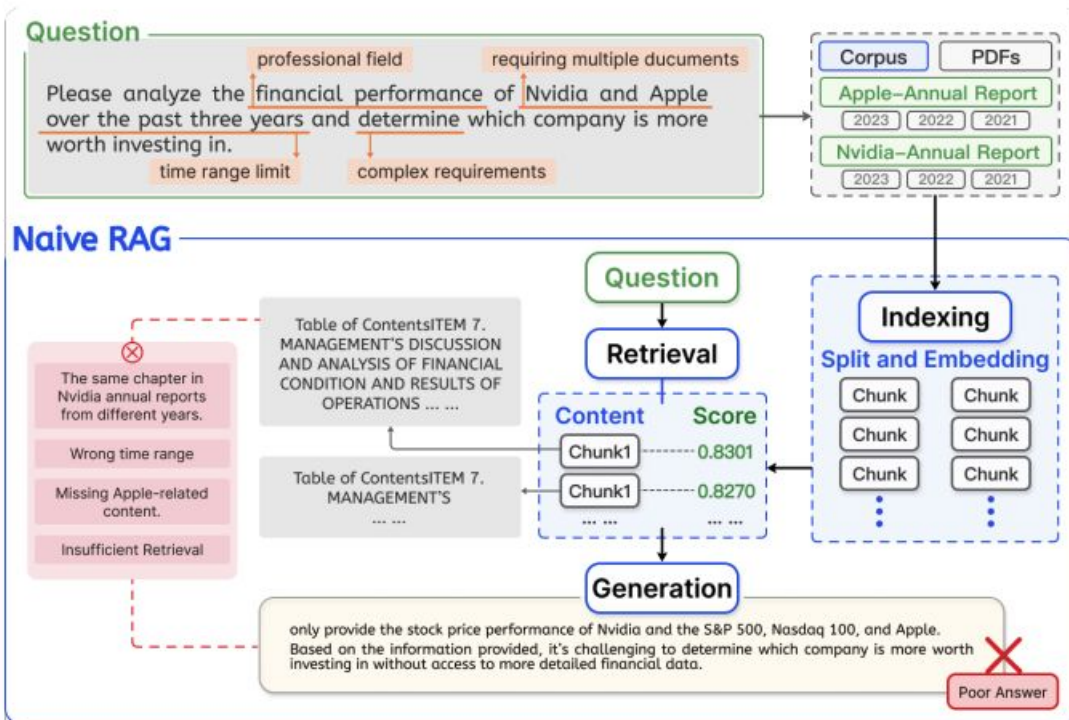


Nice. If my phone dies right after I click send, will the person still receive my message?



Sorry, I do not have that specific information

unanswerable



Always answers

even when it shouldn't. No abstention path.

Unverified retrieval

top-k returns closest, not relevant.

Flat source priority

outdated FAQ outranks current rule.

Context-blind chunking

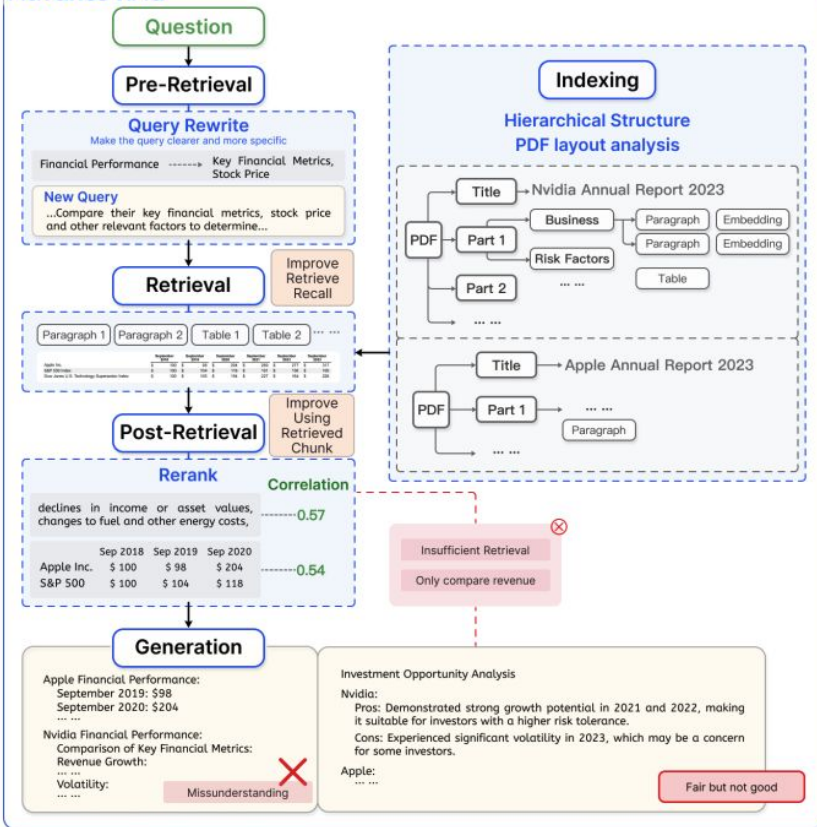
fixed splits cut meaning in half.

No safety layer

PII and sensitive content flow straight through.

A naive RAG pipeline : the PoC

Advance RAG



Smarter retrieval, same blind spots
rerank and rewrites help, ceiling stays low.

Still always answers
no path to abstain when evidence is weak.

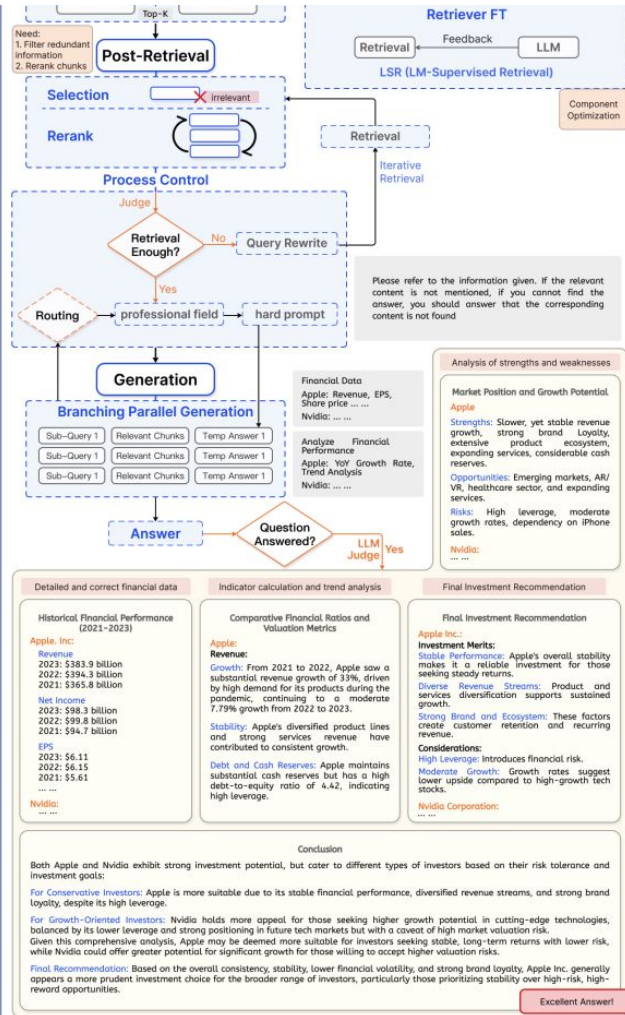
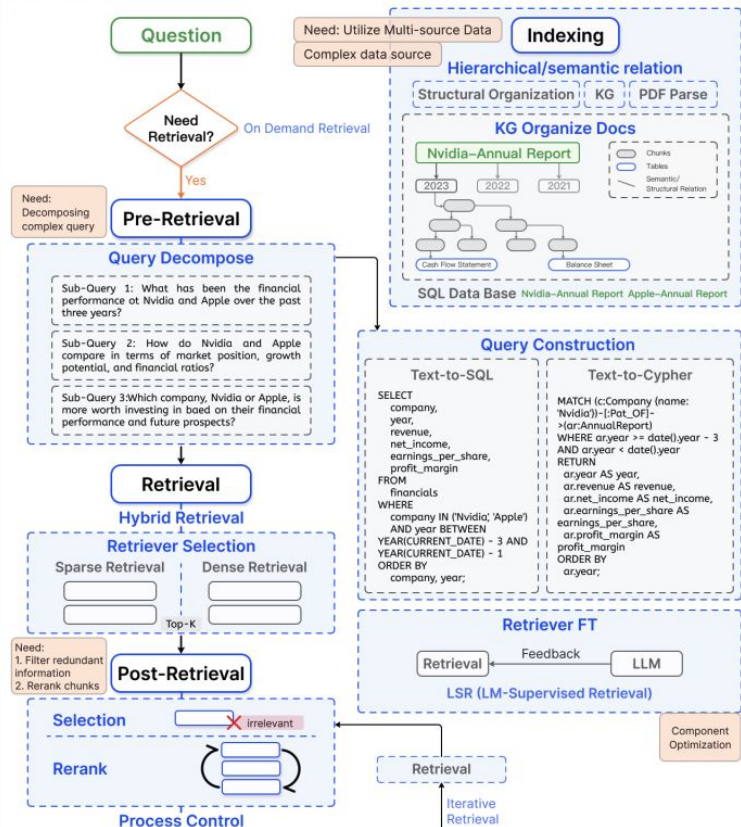
Reranking ≠ understanding
top chunk wins, even if it misses the point.

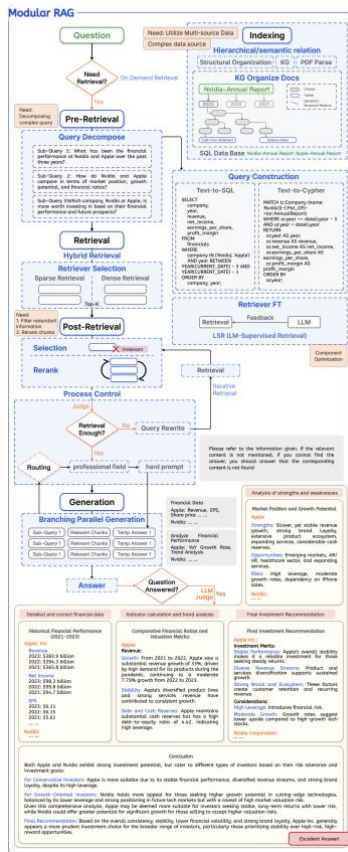
Hierarchical index, flat trust
structure improves recall, not source authority.

Linear by design
no routing, no tools, no second look.

An “advanced” RAG: optimizing the wrong thing

Modular RAG





Routes before it retrieves

different queries, different pipelines.

Trusts sources, not similarity

rules beat templates, fresh beats stale.

Calls tools when needed

SQL for structured, search for the rest.

Filters in and out

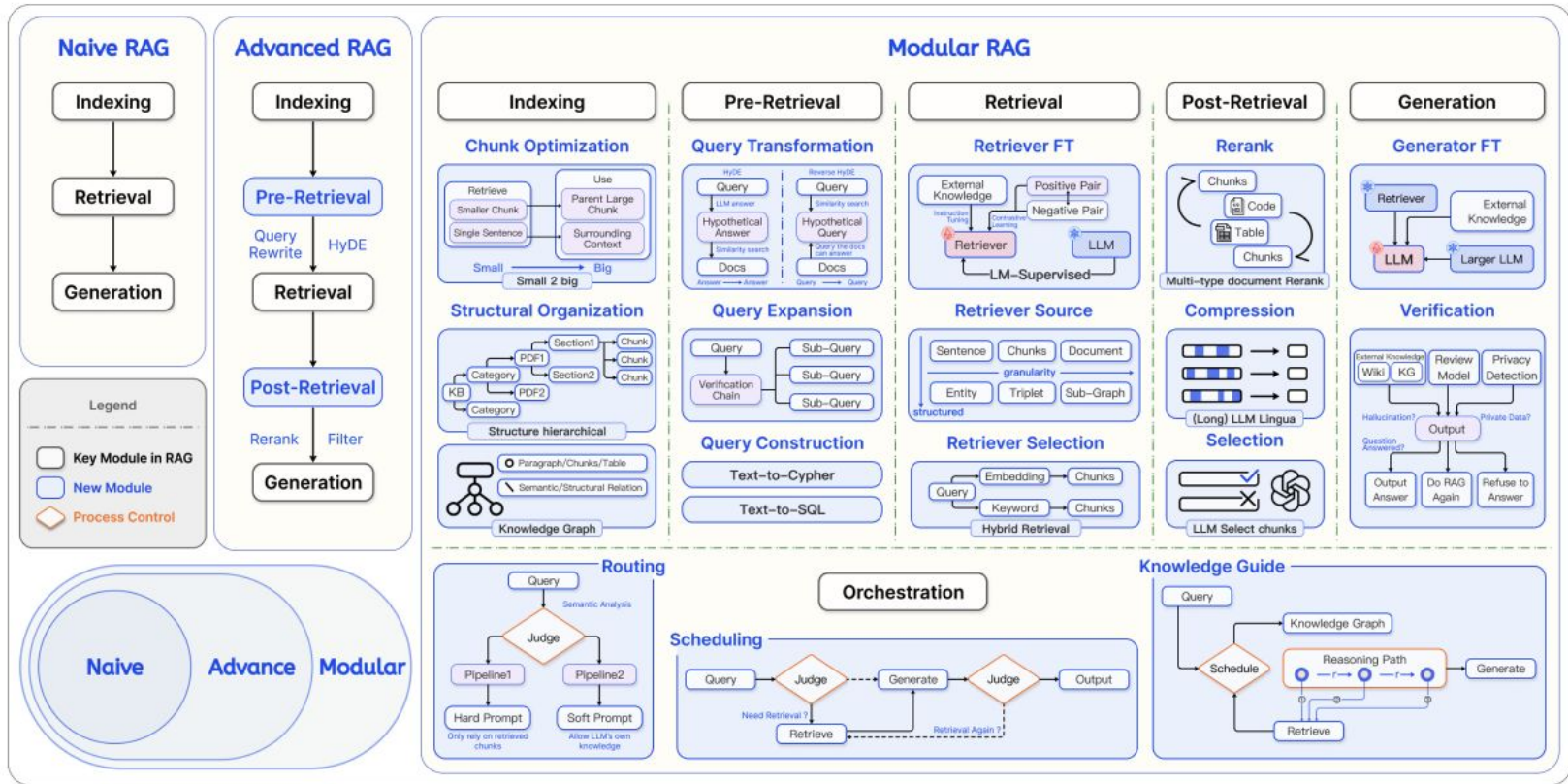
PII detection, moderation, verification.

Knows when to stay silent

no evidence, no answer.



Modular RAG: the third attempt



The 3 systems : progress

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DON'T BE SORRY X BE CAREFUL

But, it is not a miracle solution:

More moving parts

more components, more things to monitor and debug.

Latency adds up

every router, reranker, and verifier costs time.

Harder to evaluate

no single metric captures routing + retrieval + abstention.

Silence has a cost too

over-refusal frustrates users as much as wrong answers.

Not a free upgrade

needs feedback loops and tuning to actually pay off.

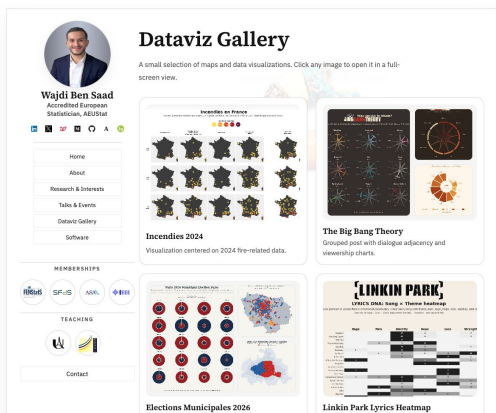
Thank You!!!



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